Gen4 Application Note

Title	Edit Title in document properties
Filename	CApp Note - Genus-HVLP-D8- using I-O ressources with I-protocol.docx
Date of Creation	18/06/2018 05:54:00 by Eric Senart
Last Updated	19/10/2018 05:56:00 by
Revision	1

Contents

1. Introduction 2
2. I-Protocol Status Message *(Lecture des états des Entrées/Sorties Digitales)* 3
3. I-Protocol Control Message *(Ecriture des états des Sorties Analogiques)* 7
4. Extras 11

Sevcon Ltd
TVTE
Gateshead
Tyne & Wear
England NE11 0QA

Tel +44 (0)191 4979000
Fax +44 (0)191 4824223
www.sevcon.com

Commercially confidential You are authorised to open and view any electronic copy we send you of this document within your organisation and to print a single copy. Otherwise the material may not in whole or in part be copied, stored electronically or communicated to third parties without the prior written agreement of Sevcon Limited.

1. Introduction

Footnotes[1].

This App note is complement to help using I/O resource on DRAGON8 and HVLP controllers over the CAN mainly.

For **PGN** definition refer to the following notes :

- App Note - I-Protocol Applied To Genus – customer
- App Note - Mapping J1939 Items to VehApp

[1] Footnote text.

2. I-Protocol Status Message *(Lecture des états des Entrées/Sorties Digitales)*

Analog and digital signal inputs and outputs are provided for switches, sensors, contactors, hydraulic valves and CAN communications. This I/O can be mapped to the CANbus for control by an external VCU.

All the control inputs and outputs such as the analogue inputs, digital inputs and motor encoder are referenced to the 12/24V logic circuit ground. The CANbus is connected to the control circuit ground.

There are 5 status messages associated with the Genus I-Protocol, namely:

UF1 – for digital input\output status for the 16 inputs and 8 outputs
UI1 – for analog input status (pin voltage) for user inputs 1-4 in Q4.8 format.
UI2 – for analog input status (pin voltage) for user inputs 5-8 in Q4.8 format.
UI3 – for analog input status (pin voltage) for user inputs 9-12 in Q4.8 format.
UI4 – for analog input status (pin voltage) for user inputs 13-16 in Q4.8 format.

In this document we will use the default PGN for illustration.

2.1. I-Protocol Status Message: Digital input/output status: UF1

Monitoring from DVTC JM windows (SA 0x71)

FF-71-14800 - UFO1	0	0x OK, 10 Fault, 11 Not Present
FF-71-14800 - UFO2	0	0x OK, 10 Fault, 11 Not Present
FF-71-14800 - UFO3	0	0x OK, 10 Fault, 11 Not Present
FF-71-14800 - UFO4	0	0x OK, 10 Fault, 11 Not Present
FF-71-14800 - UFO5	3	0x OK, 10 Fault, 11 Not Present
FF-71-14800 - UFO6	3	0x OK, 10 Fault, 11 Not Present
FF-71-14800 - UFO7	3	0x OK, 10 Fault, 11 Not Present
FF-71-14800 - UFO8	3	0x OK, 10 Fault, 11 Not Present
FF-71-14800 - UFI1	0	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI2	0	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI3	0	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI4	0	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI5	1	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI6	1	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI7	1	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI8	1	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI9	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI10	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI11	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI12	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI13	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI14	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI15	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-14800 - UFI16	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present

Time	ID	Name
1332.783530	1948FF71x	UF1
AOut_1_Status	0	0
AOut_2_Status	0	0
AOut_3_Status	2	2
AOut_4_Status	0	0
AOut_5_Status	3	3
AOut_6_Status	3	3
AOut_7_Status	3	3
AOut_8_Status	3	3
DIn_01	0	0
DIn_02	0	0
DIn_03	0	0
DIn_04	0	0
DIn_05	1	1
DIn_06	1	1
DIn_07	1	1
DIn_08	1	1
DIn_09	3	3
DIn_10	3	3
DIn_11	3	3
DIn_12	3	3
DIn_13	3	3
DIn_14	3	3
DIn_15	3	3
DIn_16	3	3

PGN : UF1						
signal	resource	DRAGON8	HVLP_a	HVLP_b	GEN5.S9	
UFI1	Din_01	PIN 18	PIN J3		A2	
UFI2	Din_02	PIN 11	PIN H3		B3	
UFI3	Din_03	PIN 03	PIN J4		D3	
UFI4	Din_04	PIN 19 **	PIN H4		D2	** : Ain_01 as Digital
UFI5	Din_05	PIN 12 **				** : Ain_02 as Digital
UFI6	Din_06	PIN 16				
UFI7	Din_07	PIN 02				
UFI8	Din_08	PIN 10				
UFO1	Dout_1_status	PIN 17			K2/	
UFO2	Dout_2_status	PIN 09			L3 /	
UFO3	Dout_3_status		PIN L3		J2 /	
UFO4	Dout_4_status		PIN M3			

Controller setting:

Index	Sub-Index	Units	Name	Value	Description
	----		I-Protocol Configuration	----	
	1	ms	Digital input transmit period	50	Configures rate at which digital input signal data is transmitted (set to zero to disable)
	2	ms	Analogue input transmit period	50	Configures rate at which analogue input signal data is transmitted (set to zero to disable)
	3		Enable checksum/counter message protection	0	
	4		Transmit message priority	6	
	5		J1939 Data Page	1	
3F01h	7		PF for message UO1	40h	
	8		PF for message UO2	41h	
	9		PF for message UF1	48h	
	10		PF for message UI1	49h	
	11		PF for message UI2	4Ah	
	12		PF for message UI3	4Bh	
	13		PF for message UI4	4Ch	

Index	Sub-Index	Units	Name	Value	Description
5800h	0		Master / Slave configuration	n/a	1 (TRUE) – NMT Master (MASTER) 0 (FALSE) – NMT Slave (SLAVE)

Index	Sub-Index	Units	Name	Value	Description
3E00h	----		J1939 Configuration	----	
	0		Number of parameters	2	
	1		Source Address	71h	Source address for data being sent using J1939 based protocols
	2		Transmit without request	1	Set to 1 to allow data to be transmitted without a J1939 request message

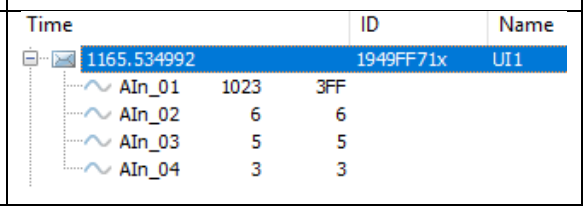
- Digital input set (Function mapping) :➔ **0x3E50** and **0x3E51**

2.2. I-Protocol Status Message: Analogue input status: UI1

(pin voltage) for user inputs 1-4

Monitoring from DVTC JM windows (SA 0x71)

FF-71-14900 - IP1	1003	application specific units
FF-71-14900 - IP2	7	application specific units
FF-71-14900 - IP3	5	application specific units
FF-71-14900 - IP4	4	application specific units



PGN : UI1						
signal	resource	DRAGON8	HVLP_a	HVLP_b	GEN5.S9	
IP1	Ain_01	PIN 16	PIN K1		E4	
IP2	Ain_02	PIN 13	PIN K2		F1	
IP3	Ain_03	PIN 04 *	PIN J1		I1/J3	* : Motor thermistor
IP4	Ain_04	PIN 02 *	PIN J2			* : Enc. A/U/Sin

Controller setting:

Index	Sub-Index	Units	Name	Value	Description
3F01h	----		I-Protocol Configuration	----	
	1	ms	Digital input transmit period	50	Configures rate at which digital input signal data is transmitted (set to zero to disable)
	2	ms	Analogue input transmit period	50	Configures rate at which analogue input signal data is transmitted (set to zero to disable)
	3		Enable checksum/counter message protection	0	
	4		Transmit message priority	6	
	5		J1939 Data Page	1	
	7		PF for message UO1	40h	
	8		PF for message UO2	41h	
	9		PF for message UF1	48h	
	10		PF for message UI1	49h	
	11		PF for message UI2	4Ah	
	12		PF for message UI3	4Bh	
	13		PF for message UI4	4Ch	

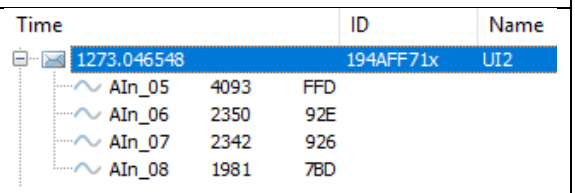
Index	Sub-Index	Units	Name	Value	Description
5800h	0		Master / Slave configuration	n/a	1 (TRUE) – NMT Master (MASTER) 0 (FALSE) – NMT Slave (SLAVE)

Index	Sub-Index	Units	Name	Value	Description
3E00h	----		J1939 Configuration	----	
	0		Number of parameters	2	
	1		Source Address	71h	Source address for data being sent using J1939 based protocols
	2		Transmit without request	1	Set to 1 to allow data to be transmitted without a J1939 request message

- Analogue input set (Function mapping) :➔ **0x3E40** and **0x3E41**

2.3. I-Protocol Status Message: Analogue input status: UI2

(pin voltage) for user inputs 5-8

Monitoring from DVTC JM windows (SA 0x71)	
	

PGN : UI2						
signal	resource	DRAGON8	HVLP_a	HVLP_b	GEN5.S9	
IP5	Ain_05		PIN H2 *			* : Motor thermistor
IP6	Ain_06					
IP7	Ain_07					
IP8	Ain_08					

Controller setting:

Index	Sub-Index	Units	Name	Value	Description
3F01h	----		I-Protocol Configuration	----	
	1	ms	Digital input transmit period	50	Configures rate at which digital input signal data is transmitted (set to zero to disable)
	2	ms	Analogue input transmit period	50	Configures rate at which analogue input signal data is transmitted (set to zero to disable)
	3		Enable checksum/counter message protection	0	
	4		Transmit message priority	6	
	5		J1939 Data Page	1	
	7		PF for message UO1	40h	
	8		PF for message UO2	41h	
	9		PF for message UF1	48h	
	10		PF for message UI1	49h	
	11		PF for message UI2	4Ah	
	12		PF for message UI3	4Bh	
	13		PF for message UI4	4Ch	

Index	Sub-Index	Units	Name	Value	Description
5800h	0		Master / Slave configuration	n/a	1 (TRUE) – NMT Master (MASTER) 0 (FALSE) – NMT Slave (SLAVE)

Index	Sub-Index	Units	Name	Value	Description
3E00h	----		J1939 Configuration	----	
	0		Number of parameters	2	
	1		Source Address	71h	Source address for data being sent using J1939 based protocols
	2		Transmit without request	1	Set to 1 to allow data to be transmitted without a J1939 request message

- Analogue input set (Function mapping) :➔ **0x3E40** and **0x3E41**

3. I-Protocol Control Message *(Ecriture des états des Sorties Analogiques)*

Analog and digital signal outputs are provided for switches, sensors, contactors, hydraulic valves and CAN communications. This I/O can be mapped to the CANbus for control by an external VCU.

3.1. I-Protocol Control Message for Analogue output: UO1

(pin voltage) for user outputs 1-4

Monitoring from DVTC JM windows (SA 0x71)		
71-71-14000 - OP1	4095	application specific units
71-71-14000 - OP2	4095	application specific units
71-71-14000 - OP3	4093	application specific units
71-71-14000 - OP4	4095	application specific units
71-71-14100 - OP5	unknown	application specific units
71-71-14100 - OP6	unknowr	application specific units
71-71-14100 - OP7	unknowr	application specific units
71-71-14100 - OP8	unknowr	application specific units

Time		ID	Name
3575.091348		19407171x	UO1
AOut_1	4095	FFF	
AOut_2	4095	FFF	
AOut_3	0	0	
AOut_4	4095	FFF	

PGN : UO1						
signal	resource	DRAGON8	HVLP_a	HVLP_b	GEN5.S9	
OP1	Aout_1	PIN 17			K2/	
OP2	Aout_2	PIN 09			L3 /	
OP3	Aout_3		PIN L3		J2 /	
OP4	Aout_4		PIN M3			

Controller setting:

Index	Sub-Index	Units	Name	Value	Description
	----		I-Protocol Configuration	----	
	1	ms	Digital input transmit period	50	Configures rate at which digital input signal data is transmitted (set to zero to disable)
	2	ms	Analogue input transmit period	50	Configures rate at which analogue input signal data is transmitted (set to zero to disable)
	3		Enable checksum/counter message protection	0	
	4		Transmit message priority	6	
	5		J1939 Data Page	1	
3F01h	7		PF for message UO1	40h	
	8		PF for message UO2	41h	
	9		PF for message UF1	48h	
	10		PF for message UI1	49h	
	11		PF for message UI2	4Ah	
	12		PF for message UI3	4Bh	
	13		PF for message UI4	4Ch	

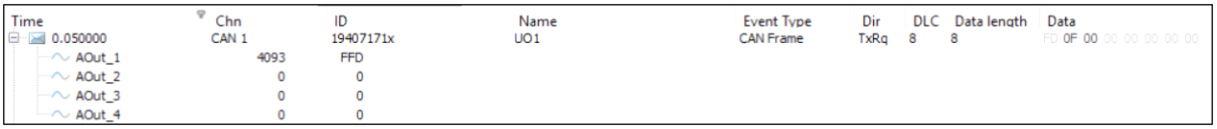
Index	Sub-Index	Units	Name	Value	Description
5800h	0		Master / Slave configuration	n/a	1 (TRUE) – NMT Master (MASTER) 0 (FALSE) – NMT Slave (SLAVE)

- Analogue output set (Function mapping) :➔ **0x3E20** and **0x3E21**

3.1.1. Controlling the ANALOGUE output :

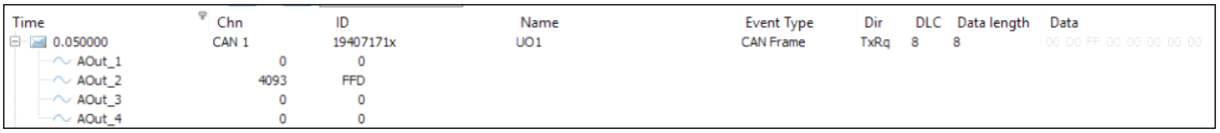
- **Analogue Output = Activated** if value between 2560 (0xA00) and 4094(0xFFE)
- **Analogue Output = disActivated** if value < 2560 (0xA00)
- **Analogue Output = Not Changed** if value = 4095 (0xFFF)

Example : Aout_01 Activated :



Time	Chn	ID	Name	Event Type	Dir	DLC	Data length	Data
0.050000	CAN 1	19407171x	UO1	CAN Frame	TxRq	8	8	FD 0F 00 00 00 00 00 00
AOut_1	4093	FFD						
AOut_2	0	0						
AOut_3	0	0						
AOut_4	0	0						

Example Aout_02 Activated :



Time	Chn	ID	Name	Event Type	Dir	DLC	Data length	Data
0.050000	CAN 1	19407171x	UO1	CAN Frame	TxRq	8	8	00 00 FF 00 00 00 00 00
AOut_1	0	0						
AOut_2	4093	FFD						
AOut_3	0	0						
AOut_4	0	0						

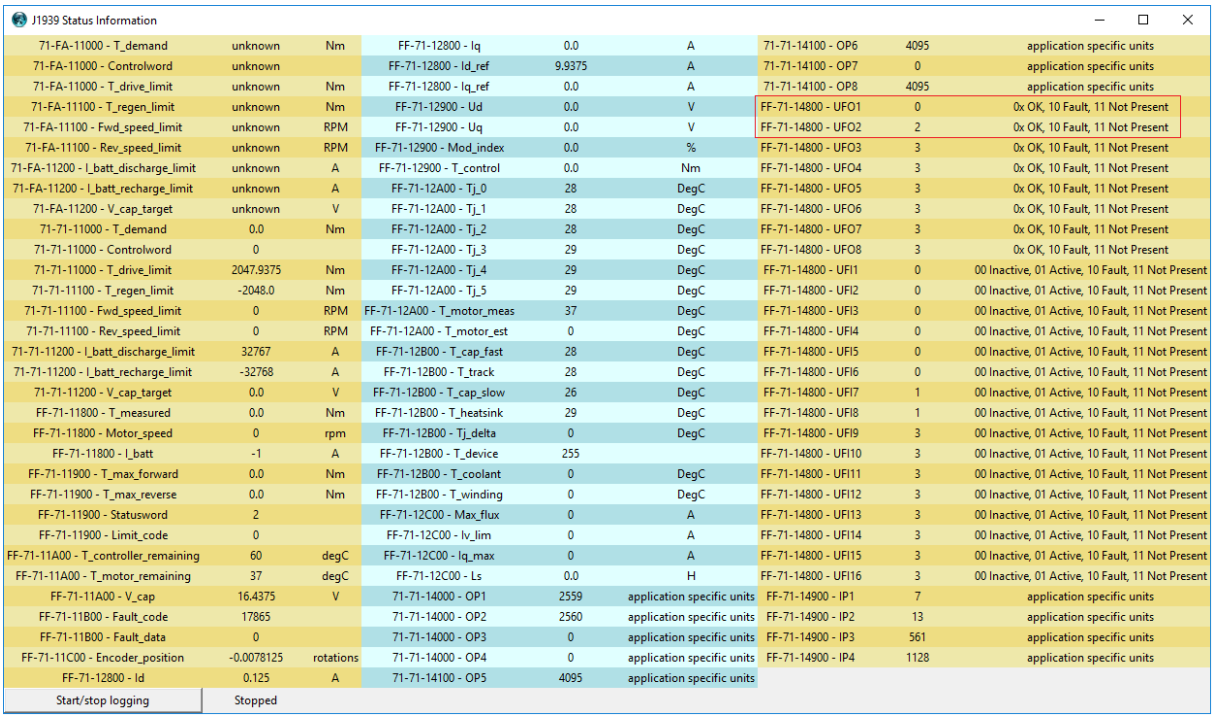
As Destination address, VCU will use the 'preferred Address' of the module supporting the resource.
Le Calculateur la valeur 'Prefered address' du module comme 'Destination Adresse' du module qui support les sorties.

3.1.2. Diagnostic of the ANALOGUE output:

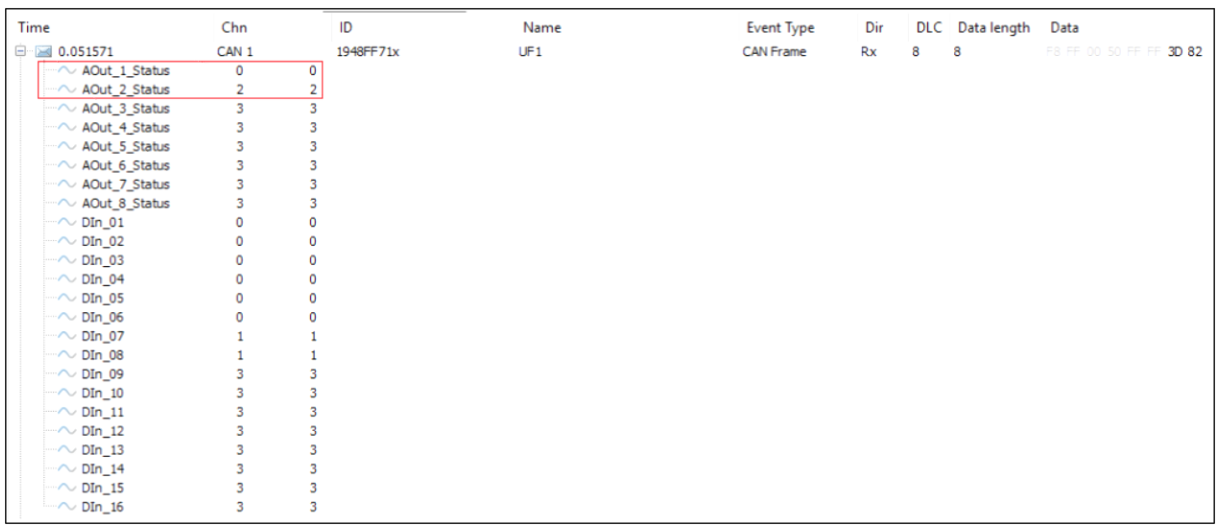
In section 2.1., **UF1** shown the Analogue outputs status !

In the following example for the DRAGON8:

- The Analogue output N°2 (Dout_2) is activated with an LED (low consumption load) → the output current is too low ➔ Dout_2 status = 2 = Fault
  - We can see the LED blinking; this is due to the periodic diagnostic test by the (as long as a fault is detected).
- If the load impedance is enough, the current consumption is higher than the minimum value ➔ Dout_2 status = 0 ➔ OK
- If there is NO load connected to the physical output ➔ status = 3 = load NO PRESENT
- According to the Analogue output driver used by the DRAGON8 the current consumption should be **> 23mA**



71-FA-11000 - T_demand	unknown	Nm	FF-71-12800 - Iq	0.0	A	71-71-14100 - OP6	4095	application specific units
71-FA-11000 - Controlword	unknown		FF-71-12800 - Id_ref	9.9375	A	71-71-14100 - OP7	0	application specific units
71-FA-11000 - T_drive_limit	unknown	Nm	FF-71-12800 - Iq_ref	0.0	A	71-71-14100 - OP8	4095	application specific units
71-FA-11100 - T_regen_limit	unknown	Nm	FF-71-12900 - Ud	0.0	V	FF-71-14800 - UFO1	0	0x OK, 10 Fault, 11 Not Present
71-FA-11100 - Fwd_speed_limit	unknown	RPM	FF-71-12900 - Uq	0.0	V	FF-71-14800 - UFO2	2	0x OK, 10 Fault, 11 Not Present
71-FA-11100 - Rev_speed_limit	unknown	RPM	FF-71-12900 - Mod_index	0.0	%	FF-71-14800 - UFO3	3	0x OK, 10 Fault, 11 Not Present
71-FA-11200 - I_batt_discharge_limit	unknown	A	FF-71-12900 - T_control	0.0	Nm	FF-71-14800 - UFO4	3	0x OK, 10 Fault, 11 Not Present
71-FA-11200 - I_batt_recharge_limit	unknown	A	FF-71-12A00 - Tj_0	28	DegC	FF-71-14800 - UFO5	3	0x OK, 10 Fault, 11 Not Present
71-FA-11200 - V_cap_target	unknown	V	FF-71-12A00 - Tj_1	28	DegC	FF-71-14800 - UFO6	3	0x OK, 10 Fault, 11 Not Present
71-71-11000 - T_demand	0.0	Nm	FF-71-12A00 - Tj_2	28	DegC	FF-71-14800 - UFO7	3	0x OK, 10 Fault, 11 Not Present
71-71-11000 - Controlword	0		FF-71-12A00 - Tj_3	29	DegC	FF-71-14800 - UFO8	3	0x OK, 10 Fault, 11 Not Present
71-71-11000 - T_drive_limit	2047.9375	Nm	FF-71-12A00 - Tj_4	29	DegC	FF-71-14800 - UFI1	0	00 Inactive, 01 Active, 10 Fault, 11 Not Present
71-71-11100 - T_regen_limit	-2048.0	Nm	FF-71-12A00 - Tj_5	29	DegC	FF-71-14800 - UFI2	0	00 Inactive, 01 Active, 10 Fault, 11 Not Present
71-71-11100 - Fwd_speed_limit	0	RPM	FF-71-12A00 - T_motor_meas	37	DegC	FF-71-14800 - UFI3	0	00 Inactive, 01 Active, 10 Fault, 11 Not Present
71-71-11100 - Rev_speed_limit	0	RPM	FF-71-12A00 - T_motor_est	0	DegC	FF-71-14800 - UFI4	0	00 Inactive, 01 Active, 10 Fault, 11 Not Present
71-71-11200 - I_batt_discharge_limit	32767	A	FF-71-12B00 - T_cap_fast	28	DegC	FF-71-14800 - UFI5	0	00 Inactive, 01 Active, 10 Fault, 11 Not Present
71-71-11200 - I_batt_recharge_limit	-32768	A	FF-71-12B00 - T_track	28	DegC	FF-71-14800 - UFI6	0	00 Inactive, 01 Active, 10 Fault, 11 Not Present
71-71-11200 - V_cap_target	0.0	V	FF-71-12B00 - T_cap_slow	26	DegC	FF-71-14800 - UFI7	1	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-11800 - T_measured	0.0	Nm	FF-71-12B00 - T_heatsink	29	DegC	FF-71-14800 - UFI8	1	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-11800 - Motor_speed	0	rpm	FF-71-12B00 - Tj_delta	0	DegC	FF-71-14800 - UFI9	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-11800 - I_batt	-1	A	FF-71-12B00 - T_device	255		FF-71-14800 - UFI10	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-11900 - T_max_forward	0.0	Nm	FF-71-12B00 - T_coolant	0	DegC	FF-71-14800 - UFI11	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-11900 - T_max_reverse	0.0	Nm	FF-71-12B00 - T_winding	0	DegC	FF-71-14800 - UFI12	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-11900 - Statusword	2		FF-71-12C00 - Max_flux	0	A	FF-71-14800 - UFI13	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-11900 - Limit_code	0		FF-71-12C00 - Iv_lim	0	A	FF-71-14800 - UFI14	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-11A00 - T_controller_remaining	60	degC	FF-71-12C00 - Iq_max	0	A	FF-71-14800 - UFI15	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-11A00 - T_motor_remaining	37	degC	FF-71-12C00 - Ls	0.0	H	FF-71-14800 - UFI16	3	00 Inactive, 01 Active, 10 Fault, 11 Not Present
FF-71-11A00 - V_cap	16.4375	V	71-71-14000 - OP1	2559	application specific units	FF-71-14900 - IP1	7	application specific units
FF-71-11B00 - Fault_code	17865		71-71-14000 - OP2	2560	application specific units	FF-71-14900 - IP2	13	application specific units
FF-71-11B00 - Fault_data	0		71-71-14000 - OP3	0	application specific units	FF-71-14900 - IP3	561	application specific units
FF-71-11C00 - Encoder_position	-0.0078125	rotations	71-71-14000 - OP4	0	application specific units	FF-71-14900 - IP4	1128	application specific units
FF-71-12800 - Id	0.125	A	71-71-14100 - OP5	4095	application specific units			
Start/stop logging	Stopped							



Time	Chn	ID	Name	Event Type	Dir	DLC	Data length	Data
0.051571	CAN 1	1948FF71x	UF1	CAN Frame	Rx	8	8	3D 82
AOut_1_Status	0	0						
AOut_2_Status	2	2						
AOut_3_Status	3	3						
AOut_4_Status	3	3						
AOut_5_Status	3	3						
AOut_6_Status	3	3						
AOut_7_Status	3	3						
AOut_8_Status	3	3						
DIn_01	0	0						
DIn_02	0	0						
DIn_03	0	0						
DIn_04	0	0						
DIn_05	0	0						
DIn_06	0	0						
DIn_07	1	1						
DIn_08	1	1						
DIn_09	3	3						
DIn_10	3	3						
DIn_11	3	3						
DIn_12	3	3						
DIn_13	3	3						
DIn_14	3	3						
DIn_15	3	3						
DIn_16	3	3						

Analogue output driver specification for DRAGON8:

7.4 Electrical Characteristics Diagnostic Function

Table 9 Electrical Characteristics: Diagnostics

V_S = 8 V to 36 V, T_J = -40 °C to +150 °C (unless otherwise specified).
Typical values are given at V_S = 28 V, T_J = 25 °C

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
Load Condition Threshold for Diagnostic							
Open load detection threshold in OFF state	V_S - $V_{OL(OFF)}$	4	–	6	V	[1] V_{IN} = 0 V V_{DEN} = 4.5 V See Figure 26	P_7.5.1
Open load detection threshold in ON state	$I_{L(OL)}$	4	–	25	mA	V_{IN} = V_{DEN} = 4.5 V $I_{IS(OL)}$ = 8 µA See Figure 24 See Figure 46	P_7.5.2

4. Extras



A NOTE indicates important information that helps you make better use of your computer system.



A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.



A WARNING indicates the potential for bodily harm and tells you how to avoid the problem.